

Part A: Multiple choice

(24 marks)

-Be sure that you have 12 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12

Q1) Suppose that the augmented matrix for a system of linear equations has been reduced by row operation to the given

$$\text{form } \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} \text{ The solution of the linear system (respectively } x, y, z) \text{ is}$$

- a) 2, -3, 2 b) 2, -3, 1 c) -1, 0, -1 d) 4, -3, 2

Q2) The values of k , if any that satisfy the equation $\begin{bmatrix} k & 1 & 1 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is. a) -2 b) 8 c) 1 d) -1

Q3) If $(I - A^T)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{-1}{3} \end{bmatrix}$, then $\text{Tr}(A) =$ a) 4 b) 8 c) -49 d) 3

Q4) For some nonsingular square matrix A , assume that $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 2 \end{bmatrix}$ the solution of the system

$$AX = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ is } \quad \text{a) } X = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} \quad \text{b) } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{c) } X = \begin{bmatrix} -1 \\ 1 \\ -4 \end{bmatrix} \quad \text{d) } X = \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$$

Q5) The system $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

- a) no solution b) two solution c) infinitely many solutions d) unique solution

Q6) The values of k for which $\det(A) = 0$, where $A = \begin{bmatrix} 7 & 0 & 0 \\ -3 & k & -6 \\ 0 & 0 & k-4 \end{bmatrix}$ is a) 4, 3 b) 1, 6 c) 4, 6 d) 1, -3

Q7) If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 4$ evaluate the $\begin{bmatrix} a & b & c \\ -2g & -2h & -2i \\ d & e & f \end{bmatrix} =$ a) -9 b) 12 c) -12 d) 8

Q8) The value of x , if any that satisfy the equation $\begin{vmatrix} 2 & x+2 \\ -1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -9 & x-2 & 0 \\ 8 & 5 & 2 \end{vmatrix}$ is. a) 12 b) 4 c) 2 d) 6

Q9) Let the vectors $v = (2, 5)$, $w = (-4, 6)$ the components of the $v+2w$ is
a) (6, 10) b) (16, 10) c) (6, -10) d) (-6, 17)

Q10) The terminal point of the vector that is equivalent to $u = (2, 4, 6)$ and whose initial point is $B(-3, -1, 2)$,
a) (-1, 3, 8) b) (8, -1, 1) c) (2, -1, 6) d) (-1, 3, 4)

Q11) Let $u = (2, 3, 0)$ and $v = (-1, 2, 2)$ the distance between u and v is a) $\sqrt{17}$ b) $\sqrt{14}$ c) $\sqrt{11}$ d) $\sqrt{10}$

Q12) Let $A = \begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ the Eigen values = a) 2, 3 b) -2, 3 c) -2, -3 d) 2, -3

Part B: Fill the blank

(15 marks)

Q1: Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ Then $|(2A)^{-1}| = \underline{\hspace{2cm}}$

Q2: Let $A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ the $\text{adj } A^{-1} = \underline{\hspace{2cm}}$

Q3: Let $V = \mathbb{R}^2$ define the operations $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + 2x_2, y_1 - 2)$ and $c \odot (x, y) = (cx, c + y)$ then $4 \odot ((2, 3) \oplus (-2, 5)) = \underline{\hspace{2cm}}$

Q4: Let $u = (0, 1, -1)$, $v = (1, 2, 0)$ then the cosine of the angle θ between u and v is $\underline{\hspace{2cm}}$

Q5: Let the vectors $u = (2, k)$ and $v = (3, 6)$, the value of k such that u and v are orthogonal $\underline{\hspace{2cm}}$

Part C: Written questions

(12 marks)

Question 1: Let $A = \begin{bmatrix} 1 & 5 & 6 & 2 \\ 2 & 4 & -2 & 1 \\ 6 & 12 & -6 & 3 \\ 1 & 0 & -1 & 2 \end{bmatrix}$ show that $\det(A) = 0$, without directly evaluating the determinant, by state the property which use.

Question 2: let $W = \{f \in p_2 : a_2 \neq 0\}$ where p_2 set of vectors for 2nd order polynomials. Show that W not subspace of the vector space.

Question 3: Let $p_1 = 1 - x$, $p_2 = 5 + 3x - 2x^2$, $p_3 = 1 + 3x - x^2$ are p_1, p_2, p_3 linearly dependent or independent? Why.

Question 4: Let $\det(A) = 2$, $\det(B) = 4$ find $\det(A^{-1}B)$.

With my best wishes ,

- 1 → d
- 2 → d
- 3 → d
- 4 → A
- 5 → A
- 6 → C
- 7 → d
- 8 → A
- 9 → d
- 10 → A
- 11 → B
- 12 → D

part B

$$1 = \frac{-1}{96}$$

$$2 - \begin{bmatrix} 1 & \frac{3}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$3) (-8, 5)$$

$$4) \theta = \cos^{-1} \frac{2}{\sqrt{10}}$$

$$5) K = -1$$

part C

$$1 - R_2 \leftarrow 3R_2$$

3: linearly dependent
because the det = zero

4) det A = 2 det B = 4
the find det(A⁻¹B)

$$= \det A^{-1} \times \det B$$

$$\frac{1}{\det A} \times \det B = \frac{1}{2} \times 4 = 2$$

No.

Date

Subject